HW #9

1. (a) $x(n) = \delta(n-5)$ where $N > 5$

DTFT: $X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j n\Omega} = \sum_{n=-\infty}^{\infty} \delta(n-5) e^{-jn\Omega} = e^{-j5\Omega}$

DFT: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} \delta(n-5) e^{-j\frac{2\pi kn}{N}} = e^{-j\frac{2\pi k \cdot 5}{N}} = e^{-j\frac{10\pi \cdot k}{N}}$

Refer the plots from the attachment.

(b) $x(n) = (-1)^n u(n)$

DTFT: $X(\Omega) = \sum_{n=-\infty}^{\infty} (-1)^n u(n) e^{-j\Omega n} = \sum_{n=0}^{\infty} (-1)^n e^{-jn\Omega} = \sum_{n=0}^{\infty} (-e^{-j\Omega})^n$

since $q = e^{-j\Omega}$ and $|q| = 1$, so that the summation does not converge and the DTFT does not exist

DFT: $X(k) = \sum_{n=0}^{N-1} (-1)^n u(n) e^{-j\frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} (-e^{-j\frac{2\pi k}{N}})^n = \frac{1-(-e^{-j\frac{2\pi kn}{N}})^N}{1+e^{-j\frac{2\pi k}{N}}} = \frac{1-(-1)^N}{1+e^{-j\frac{2\pi k}{N}}}$

(c) $x(n) = 4^{-n} u(n)$

DTFT: $X(\Omega) = \sum_{n=-\infty}^{\infty} 4^{-n} u(n) e^{-j\Omega n} = \sum_{n=0}^{\infty} 4^{-n} e^{-j\Omega n} = \sum_{n=0}^{\infty} (\frac{1}{4} e^{-j\Omega})^n = \frac{1}{1-\frac{1}{4}e^{-j\Omega}} = \frac{4}{4-e^{-j\Omega}}$

DFT: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} (4^{-n}) e^{-j\frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} (\frac{1}{4} e^{-j\frac{2\pi k}{N}})^n = \frac{1-(\frac{1}{4}e^{-j\frac{2\pi k}{N}})^N}{1-\frac{1}{4}e^{-j\frac{2\pi k}{N}}} = \frac{1-(\frac{1}{4})^N}{1-\frac{1}{4}e^{-j\frac{2\pi k}{N}}}$

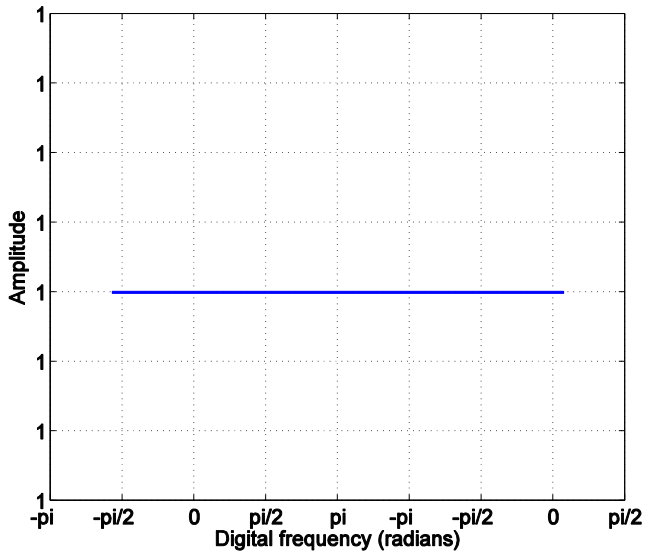
(d): $x(n) = \{00111|00\}$

DTFT: $X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = e^{-j\Omega 2} + e^{-j\Omega 3} + e^{-j\Omega 4} + e^{-j\Omega 5}$

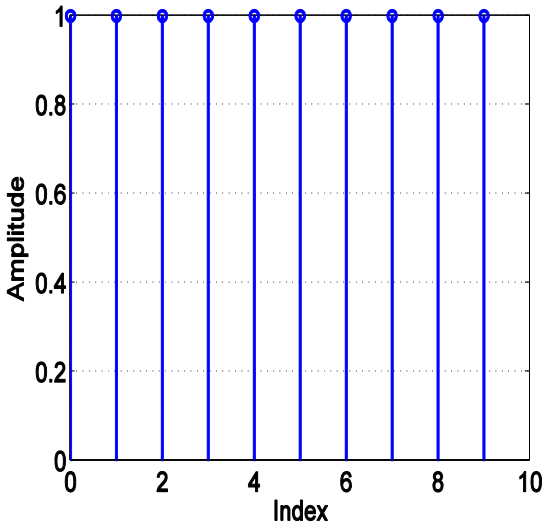
DFT: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} = e^{-j\frac{2\pi k \cdot 2}{N}} + e^{-j\frac{2\pi k \cdot 3}{N}} + e^{-j\frac{2\pi k \cdot 4}{N}} + e^{-j\frac{2\pi k \cdot 5}{N}}$

$= e^{-j\frac{4\pi k}{N}} + e^{-j\frac{6\pi k}{N}} \quad e^{-j\frac{8\pi k}{N}} + e^{-j\frac{10\pi k}{N}}$

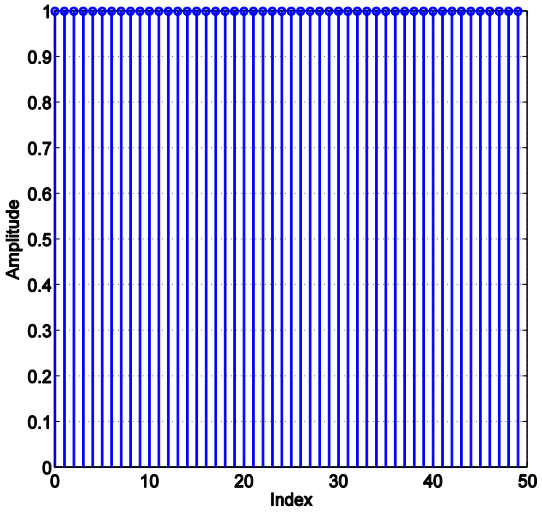
(a) DTFT



DFT with N=10



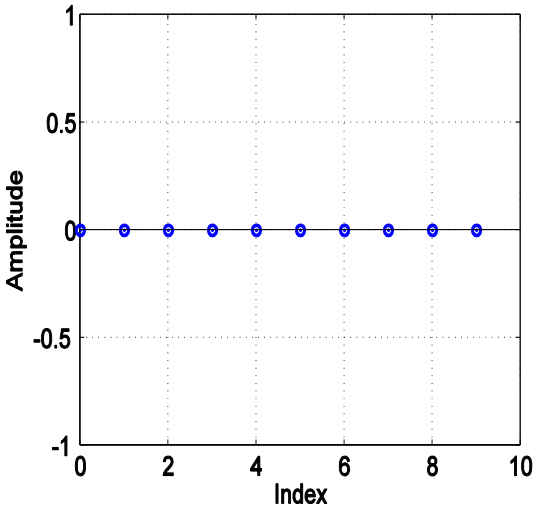
DFT with N=50



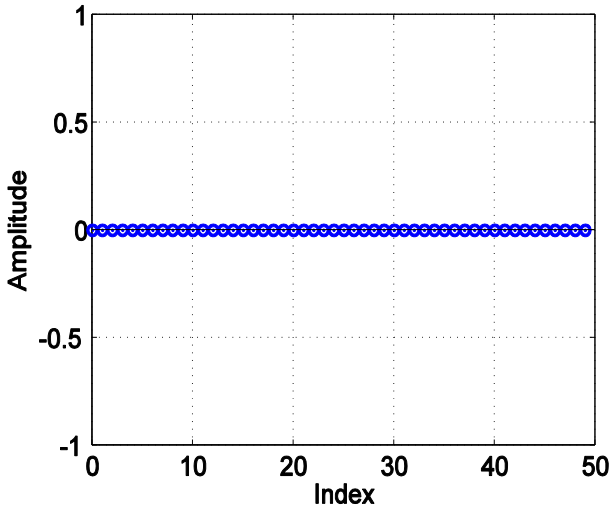
(b) DTFT

Does not exist

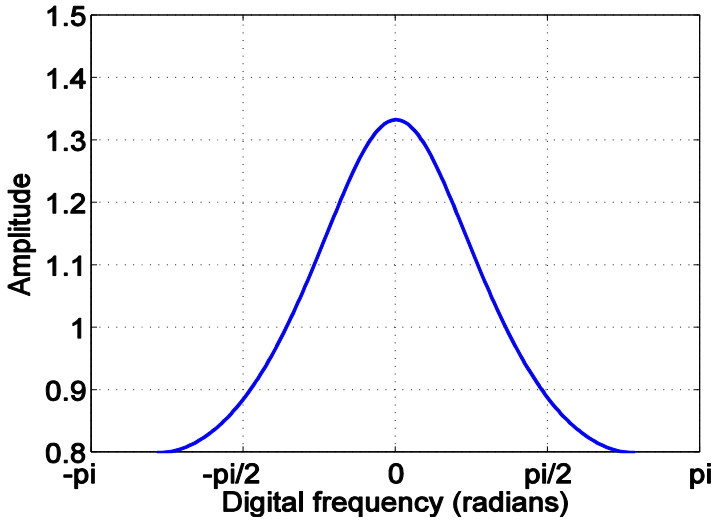
DFT with N=10



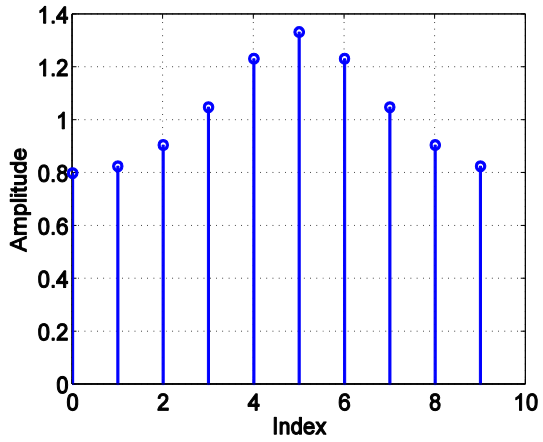
DFT with N=50



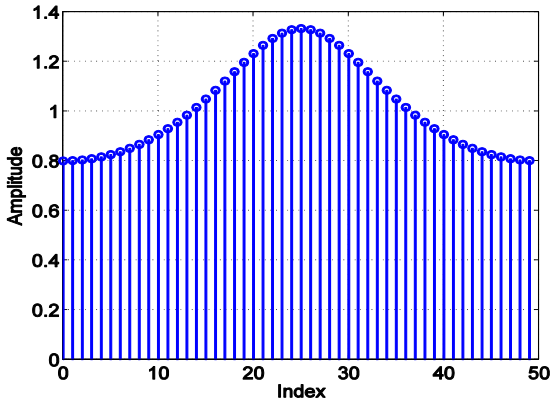
(c) DTFT



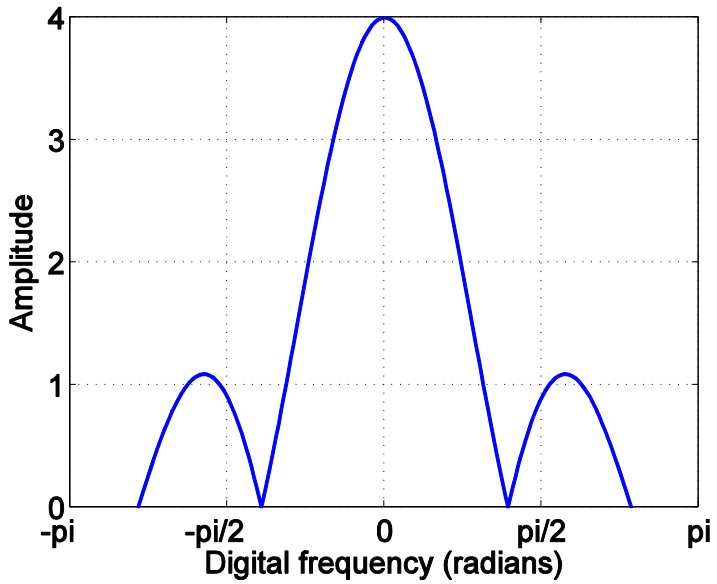
DFT with N=10



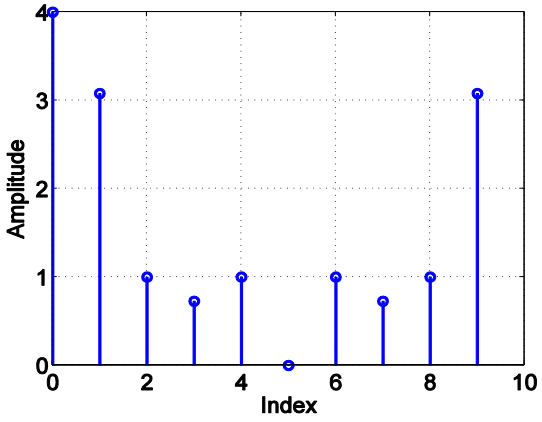
DFT with N=50



(d) DTFT



DFT with N=10



DFT with N=50

